Synthetic Relativity: Tri-modality and the Fixpoint Semantics of the Logical Observer

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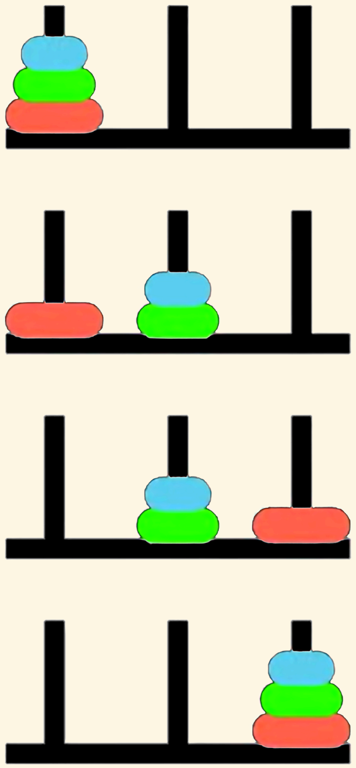
The situation calculus

The situation calculus (Towers of Hanoi)



Fluents: $OnPeg(d, p, s)$

Initial Situation:

$$OnPeg(D_1, Peg_1, S_0) \wedge$$
$$OnPeg(D_2, Peg_1, S_0) \wedge$$
$$OnPeg(D_3, Peg_1, S_0)$$

Actions: $move(d, p_{from}, p_{to})$

Goal: All disks on Peg_3.

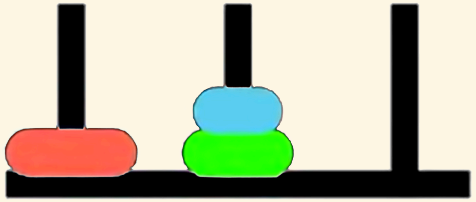
Reiter's Induction Axiom

The Foundational Induction Axiom: Second Order

$$\forall P.\ [P(S_0) \land \forall a, s\,(P(s) \to P(do(a, s)))] \to \forall s\, P(s)$$

- Every valid situation is mathematically guaranteed to be a finite sequence of actions rooted at S_0.
- Establishes the situation domain as an **inductive** datatype.
- Note that standard first order logic cannot enforce reachability due to the Compactness Theorem.

Situations & Actions



$$
\begin{aligned}
\textit{Initial situation}: &\ S_0 \\
&\ do(move(D_1, Peg_1, Peg_3), S_0) \\
do(move(D_2, Peg_1, Peg_2), &\ do(move(D_1, Peg_1, Peg_3), S_0)) \\
do(move(D_1, Peg_3, Peg_2), do(move(D_2, Peg_1, Peg_2), &\ do(move(D_1, Peg_1, Peg_3), S_0)) \\
&\ \ldots
\end{aligned}
$$

Situations as a Least Fixpoint

Let $\mathcal{A}$ be actions and $\mathcal{S}$ be the universe of situations. Define a monotonic operator Γ:

$$\Gamma(X) = \{S_0\} \cup \{do(a, s) \mid a \in \mathcal{A}, s \in X\}$$

From the **Second-Order Induction Axiom**:

- **The Premise:** $P(S_0) \wedge \forall a, s\,(P(s) \to P(do(a, s)))$ translates to $\Gamma(P) \subseteq P$. Here, P is a **pre-fixed point** *(an over-approximation that obeys the rules, but may contain unreachable "junk" like ghost situations or disconnected time loops).*
- **The Consequent:** $\forall s\, P(s)$ asserts that the true universe of situations $\mathcal{S}$ is a subset of P.
- The axiom combines these, stating $\mathcal{S}$ is a subset of *every* valid pre-fixed point. Taking their intersection mathematically acts as a sieve, filtering out all the unshared, unreachable junk.

Situations as a Least Fixpoint (continued)

The Knaster-Tarski Theorem: least fixpoint

The intersection of all pre-fixed points of a monotonic operator yields its **least fixpoint**.

Therefore, the Second-Order Induction Axiom rigorously enforces

$$\mathcal{S} = \mu X.\,\Gamma(X)$$

Leaving only the pure, minimal valid tree.

Syntactic transformation to First Order Logic

Basic Action Theories [Lin and Reiter 94] define Theory

$$\mathcal{D} = \Sigma \cup \mathcal{D}_{S_0} \cup \mathcal{D}_{Poss} \cup \mathcal{D}_{SSA} \cup \mathcal{D}_{UNA}$$

- $\mathcal{D}_{poss}$: axioms describing the prerequisites of the primitive actions, of the form:

  $$Poss(A(\vec{x}), s) \equiv \Theta_A(\vec{x}, s),$$

  one for each primitive action A.

$$\begin{aligned} Poss(move(d, p_{from}, p_{to}), s) \equiv\ & Disk(d) \wedge (p_{from} \neq p_{to}) \wedge OnPeg(d, p_{from}, s) \wedge \\ & \neg \exists d' \Big(Disk(d') \wedge Smaller(d', d) \wedge \\ & (OnPeg(d', p_{from}, s) \vee OnPeg(d', p_{to}, s)) \Big) \end{aligned}$$

A solution to the frame problem (sometimes) [Reiter 91]

- $\mathcal{D}_{ssa}$: axioms describing the effects and non-effects of actions, of the form $F(\vec{x}, do(a,s)) \equiv \psi_F(\vec{x}, a, s)$, one for each fluent F.

1. Positive effect axioms:

$$\exists p_{from}\ (a = move(d, p_{from}, p)) \supset OnPeg(d, p, do(a,s))$$

2. Negative effect axioms:

$$\exists p_{to}\ (a = move(d, p, p_{to})) \supset \neg OnPeg(d, p, do(a,s))$$

3. Frame axioms:

$$OnPeg(d,p,s) \wedge \neg\exists p_{to}\ (a = move(d, p, p_{to})) \supset OnPeg(d, p, do(a,s))$$

4. Completeness assumption (explanation closure):

$$\neg OnPeg(d,p,s) \wedge OnPeg(d,p,do(a,s)) \supset \exists p_{from}\ (a = move(d, p_{from}, p))$$

$\mathcal{D}_{ssa} =$ positive effect axioms $+$ negative effect axioms $+$ frame axioms $+$ completeness assumption

Successor State Axioms

Suppose that for fluent $F(\vec{x}, s)$:
$\gamma_F^+(\vec{x}, a, s)$ encodes all *positive* effects; and
$\gamma_F^-(\vec{x}, a, s)$ encodes all *negative* effects.
The successor state axiom solution has the following form:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee F(\vec{x}, s) \wedge \neg\gamma_F^-(\vec{x}, a, s)$$

In our example:

$$\begin{aligned}
\gamma_{OnPeg}^+(d, p, a, s) &\stackrel{\text{def}}{=} \exists p_{from}\ (a = move(d, p_{from}, p)) \\
\gamma_{OnPeg}^-(d, p, a, s) &\stackrel{\text{def}}{=} \exists p_{to}\ (a = move(d, p, p_{to}))
\end{aligned}$$

So successor state axiom is:

$$\begin{aligned}
OnPeg(d, p, do(a, s)) \equiv {} & [\exists p_{from}\ (a = move(d, p_{from}, p))] \vee \\
& OnPeg(d, p, s) \wedge \neg[\exists p_{to}\ (a = move(d, p, p_{to}))]
\end{aligned}$$

Regression

The **regression operator** ($\mathcal{R}$) evaluates a query about a future state by pushing it backward in time.

$$\mathcal{R}[\boldsymbol{F(\vec{x}, do(a,s))}] \stackrel{\text{def}}{=} \gamma_F^+(\vec{x}, a, s) \vee \boldsymbol{F(\vec{x}, s)} \wedge \neg\gamma_F^-(\vec{x}, a, s)$$

For Example: is D_2 on Peg_2 after we move D_1 from Peg_1 to Peg_2?

1. Apply $\mathcal{R}$ to the query:

$$\mathcal{R}[\,OnPeg(D_2, Peg_3, do(move(D_1, Peg_1, Peg_2), S_0))\,]$$

2. Syntactic substitution (using our SSA):

$$\equiv [\exists p_{from}\ (move(D_1, Peg_1, Peg_2) = move(D_2, p_{from}, Peg_3))] \vee$$
$$OnPeg(D_2, Peg_3, S_0) \wedge \neg[\exists p_{to}\ (move(D_1, Peg_1, Peg_2) = move(D_2, Peg_3, p_{to}))]$$

3. Evaluate using Foundational Axioms ($\mathcal{D}_{una}$): Because our Unique Names Axioms explicitly state $D_1 \neq D_2$, both action equivalences mathematically evaluate to $False$:

$$\equiv \text{False} \vee OnPeg(D_2, Peg_C, S_0) \wedge \neg\text{False}$$

4. Logical Result:

$$\equiv OnPeg(D_2, Peg_C, S_0)$$

Foundational Limitations of Least Fixpoints?

While classical induction successfully builds a valid history, enforcing this rigid geometry creates severe structural limits for dynamic, self-reflective systems:

- **Strict Tree Topology (Unique Predecessors):** Distinct action histories never converge. Even if two different paths result in the same state, the logic traps them in completely isolated parallel universes.
- **Total Ordering (The Self-Reflection Barrier):** The strictly linear sequence of actions restricts true partial ordering and concurrency. The logic cannot step “outside” the sequence to self-reflect or re-evaluate its overarching reference frame.
- **No Native State Equivalence:** Because states are strictly defined by their rigid *syntactic history* rather than their *observable behaviour*, the topology cannot naturally “fold” or quotient identical states together.
- **The Finiteness Barrier:** By mathematical definition, a least fixpoint (μ) only constructs strictly *finite* chains. [It is fundamentally blind to continuous processes, nn

Property Persistence as a Greatest Fixpoint

Property persistence in the situation calculus captures a condition ϕ that holds *now* and is preserved after *any legal action* via regression ($\mathcal{R}$).

[Kelly & Pearce AIJ 2010] define this as the **Persistence operator** ($\mathcal{P}$):

$$\mathcal{P}(Z) \equiv \phi(s) \wedge \forall a\big(Poss(a,s) \rightarrow \mathcal{R}[\,Z(do(a,s))\,]\big)$$

Iteratively applying this meta-level calculation computes the maximal invariant without explicitly invoking the second-order induction axiom.

- By wrapping the single-step regression operator inside a monotonic formula transformer ($\mathcal{P}$), the future can be evaluated back in the current state.

The Syntactic Limit: Greatest Fixpoints (νZ)

- Property persistence is an infinite-horizon **safety property**.
- The persistence condition is the **greatest fixpoint** of this operator: $\nu Z.\ \mathcal{P}(Z)$.
- *The Problem:* Regression builds finite states. To syntactically evaluate this infinite fixpoint, we must shift our mathematical framework.

Category Theory

The Dual Problems: Algebras & Coalgebras

To transition from finite histories to observing infinite Greatest Fixpoints, we harness **Category Theory**. An *endofunctor* (F) over a state space (S) is a structural template that shapes the resulting data object, $F(S)$.

- **Algebra (Situation Calculus / Constructor):** Morphism arrows point *inward*: $F(S) \to S$
  - Our endofunctor shapes the inputs: $F(S) = Action \times S$
  - The *morphism* evaluates to: $(Action \times State_{current}) \to State_{next}$
  - *Concept:* *Construct* a history from the bottom-up.
- **Coalgebra (Property Persistence / Observer):** Morphism arrows point *outward*: $S \to F(S)$
  - Our endofunctor shapes the emitted data: $F(S) = Observation \times S$
  - The morphism evaluates to: $State_{current} \to (Observation \times State_{next})$
  - *Concept:* Take a running state, break off an *observation*, and unfold the remaining state into the future.

Categorical Proof Techniques: Coinduction & Stone Duality

To reason about infinite streams when operating in coalgebras, we harness two categorical proof mechanisms.

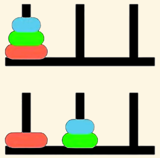
1. Coinduction (The Dual of Induction) While standard induction proves a finite base case and builds upward, coinduction works top-down. It *proposes* a Greatest Fixpoint (an infinite rule, νZ) and verifies it locally.

If the observation holds right now, and the rules guarantee it holds in the next step, the Greatest Fixpoint is guaranteed to hold forever.

2. Stone Duality (Syntax-Semantics Mirror) In Category Theory, **Stone Duality** dictates an inverse mirror-image between **Syntax** (logical formulas) and **Semantics** (topological state space).

- *The Syntactic Transformation:* If our logic formally declares two distinct observational paths to be identical (Bisimulation Equivalence) . . .
- . . . *The Semantic Result:* Stone Duality guarantees the underlying semantic space automatically folds (quotients) to make those paths topologically identical.

Example: Coinduction in Towers of Hanoi



IF we observe the *relative movement* of the smallest disk D_1 (alternating moves with D_2 or D_3) on $OnPeg(d, p, s)$:

$$\Delta p \equiv (p_{to} - p_{from}) \pmod 3$$

1. The Coalgebraic Stream: Solving the puzzle generates an infinite stream of spatial jumps:

$$\langle +2 \rangle, \langle -1 \rangle, \langle -1 \rangle, \langle +2 \rangle \dots$$

2. The Coinductive Proposal: We propose an idealised Greatest Fixpoint where the disk *always* cycles left:

$$ParityInvariant \equiv \nu Z.\ \langle -1 \rangle Z$$

Example: Coinduction in Towers of Hanoi (Continued)

3. The Coinductive Step (Bisimulation): Following $+2$, the stream outputs a jump of -1. How does our *proposed invariant* survive this paradox?

To satisfy the local coinductive check, the logic evaluates the domain's geometry. It formally recognises that jumping right twice $(+2)$ is structurally indistinguishable from jumping left once (-1). It applies this coinductive equivalence:

$$\forall \phi.\ \langle +2 \rangle \phi \ \equiv \ \langle -1 \rangle \phi$$

4. The Stone Duality Resolution: By syntactically equating these propositions, **Stone Duality** automatically forces the absolute state space to fold into a modulo-3 *quotient ring* $(\mathbb{Z}_3)$!

This is our new syntactic transformation!

Synthetic relativity axioms

Tri-modality & ternary representation

Logical entailment operates within a **Total Category**—a mathematically complete Stone Duality glues syntax and semantics together.

It is evaluated relative to a specific baseline **Reference Frame**, S_{f}. This facilitates the rules and symmetries over which the observer's duality is constructed.

- It utilises a ternary structure, drawing from **Routley-Meyer semantics** for substructural, relevance, and intuitionistic logics, involving relations $R(x, y, \mathrm{S}_{\mathrm{f}})$.
- This allows us to decouple the synthetic (axiomatic) transformations from the topological space.

Synthetic Relativity Axioms

The space is governed by three interacting modalities.

- RDo **(The Subproblem / Syntax): The Inductive Accumulator ($\Box_1$)** Operates in constructive, non-Boolean logic. Recursively gathers finite, computationally observable properties (compact elements) along a directed logical path.
- ODo **(The Master Problem / Semantics): The Limit/Creator Modality ($\Box_2$)** The semantic evaluator and *interval invariant*.

Performs topological closure—evaluating the directed set generated by RDo *and snapping it into a discrete, continuous limit.*

- PDo **(Autonomous Progression): The Dynamic Transition Functor ($\Box_3$)** The transition modality that governs the forward evolution of the system.

Mathematically, it acts as an endofunctor within the Total Category, orchestrating the step-by-step recursive interaction (categorical adjunction) of local syntax (RDo) *and global semantics* (ODo) *relative to* S_f.

Constructive Accumulation

Axiom 1: The Axiom of Constructive Accumulation

The Dynamic Transition Functor (PDo) governs the autonomous progression of the system via a strict recursive transition law relative to the Reference Frame (S_f):

- **Base Case (Initialisation):** Let $S_r^{(0)}$ and $S_o^{(0)}$ be the initial states.

$$\Sigma_0 = \mathrm{PDo}\big(S_r^{(0)},\ S_o^{(0)},\ S_f\big)$$

- **Successor Rule (Inductive Step):** For any step $n \geq 0$, the next state strictly nests new inductive syntax (RDo) and semantic coinduction (ODo):

$$\Sigma_{n+1} = \mathrm{PDo}\Big(\mathrm{RDo}\big(R_{n+1}, S_r^{(n)}\big),\ \mathrm{ODo}\big(O_{n+1}, S_o^{(n)}\big),\ S_f\Big)$$

Constructive Accumulation (continued)

Unrolling this recursive law mathematically generates an endlessly nested trace of accumulating evidence:

$$\mathrm{PDo}(S_r, S_o, S_f)$$

$$\mathrm{PDo}\big(\,\mathrm{RDo}(R_1, S_r),\ \mathrm{ODo}(O_1, S_o),\ S_f\,\big)$$

$$\mathrm{PDo}\Big(\,\mathrm{RDo}\big(R_2, \mathrm{RDo}(R_1, S_r)\big),\ \mathrm{ODo}\big(O_2, \mathrm{ODo}(O_1, S_o)\big),\ S_f\,\Big)\ldots \textit{etc.}$$

It reads: *"Relative to reference frame S_f, the accumulation of inductive evidence* RDo *yields the coinductively realised semantic limit* ODo*."*

Semantic Realisation

Theorem I: Semantic Realisation

Semantic Realisation is the mathematical state where syntax and semantics converge. Because the logic builds upon finite, verifiable observables, autonomous progression halts when the topology reaches Algebraic Compactness:

$$\mu(\mathrm{PDo}) = \mathrm{ODo}\left(\bigsqcup_{i\in R} \mathrm{RDo}_i(\perp)\right)$$

- $\bigsqcup \mathrm{RDo}$ **(Syntax):** The builder generates a raw, directed chain of evidence from the ground up. A syntax string alone has no inherent "meaning."
- ODo **(Semantics):** The top-down evaluator reads the syntax trace and evaluates its truth value.

Proof Sketch: Corresponds to the computation of the **Kleene Fixed-Point Theorem**.

Semantic Realisation (continued)

- The ultimate stable fixpoint (μ) is caught when ODo evaluates the limit (Supremum, $\bigsqcup$) of the recursively enumerable inductive proofs (RDo).
- This accumulation mathematically begins from an initial state of ignorance ($\perp$), but is constructed and evaluated strictly relative to the rules of the **reference frame** (**S_f**).
- This mathematically realised limit acts as the interval semantic invariant for that specific frame.

The Synthetic Transformation

Axiom 2: The Axiom of Synthetic Transformation

Once Realisation is achieved, the relativistic reference frame mathematically shifts. The categorically preserved invariant (the semantic limit, O) becomes the new reference frame (S_f) for the next order of progression.

$$\mathrm{PDo}(N, O, S_f) \longrightarrow \mathrm{PDo}(X, Y, O)$$

- **Left Side:** The realised state computed within the initial reference frame (S_f).
- **The Arrow ($\longrightarrow$):** The Synthetic Transformation (Topological closure and categorical Change of Base).
- **Right Side:** A new, higher-order autonomous progression begins.

The semantic limit O has mathematically synthesised into an syntactic, discrete baseline coordinate for the new subproblem (X).

Towers of Hanoi example

1. Local Unrolling (Building the Situation Trace): The logic recursively accumulates primitive actions (moving disks), generating the topological equivalent of a nested $do()$ sequence:

$$\mathrm{PDo}\Big(\mathrm{RDo}\big(\mathbf{move}(D_1, P_2), \ldots S_f\big),\ \mathrm{ODo}\big(O_{\mathrm{target}}, \mathrm{Goal}\big),\ S_f\Big)$$

2. Limit Catching: Through top-down semantic reflection ($\Box_2$), the ODo modality mathematically coinduces a structural property of the optimal traces: The Parity Invariant (O_{parity}).

3. The Synthetic Transformation (Change of Base): The Transformation Axiom fires, executing a categorical Change of Base:

$$\mathrm{PDo}(N_{\mathrm{trace}}, O_{\mathrm{parity}}, S_f) \longrightarrow \mathrm{PDo}(X, Y, \mathbf{O}_{\mathrm{parity}})$$

The invariant (cycle D_1, alternate move, cycle D_1) emerges as the unique, strictly positive proof path through the state space.

- The new parity invariant shifts to become the Reference Frame. By adopting this rule as a **lifted axiom**, *the search tree branching factor strictly collapses from 3 down to 1.*

The Dual Fixpoints

Unifying Reachability (μ) & Persistence (ν): By intertwining syntax (RDo) and semantics (ODo), the dynamic transition functor (PDo) mathematically fuses two distinct forms of logical progression:

1. **Induction / Least Fixpoint (μ):** Governed by the RDo modality. The bottom-up synthesis of finite evidence from baseline ignorance ($\perp$). It guarantees computational *Reachability*.
2. **Coinduction / Greatest Fixpoint (ν):** Governed by the ODo modality. The top-down evaluation of continuous semantics. It guarantees *Safety and Property Persistence*.

Theorem II: Algebraic Compactness

Because **Theorem I (Semantic Realisation)** formally computes a categorical *bilimit*, the system achieves algebraic compactness. At convergence, the initial algebra of the dynamic transition functor coincides with its final coalgebra:

$$\mu(\mathrm{PDo}) \cong \nu(\mathrm{PDo})$$

The active inductive trace (μ) strictly fulfills the persistent semantic invariant (ν).

Relativistic Domain Theory in Logical Form (R-DTLF)

DTLF Logical Primitives—No negation or implication

We ground our system in Abramsky's **DTLF** (1991) based on an **intuitionistic** construction of truth, since we require a logic that fuses classical (inductive) logic with intuitionistic (coinductive) logic. DTLF formulas represent observable properties as they unfold, and because the system can only assert what it can finitely verify, the baseline syntax is strictly positive.

$$\phi, \psi ::= \top \mid \bot \mid \phi \wedge \psi \mid \phi \vee \psi \mid \mathrm{Constructors}(\phi)$$

- X, Y **(Variables):** Placeholders used to define recursive logical properties.
- $\top$ **(True / Top):** The trivial observation (a valid transition occurred). $\bot$ **(False / Falsum):** The null observation (a paradox occurred)—The baseline ignorance from which new inductive proofs begin.
- $\phi \wedge \psi$ / $\phi \vee \psi$ **(Conjunction / Disjunction):** "I simultaneously observe both ϕ and ψ" / "I observe either ϕ or ψ."
- $\langle\delta\rangle\phi$ **(The Coalgebraic Transition Modality):** This is the literal "arrow breaking in two." It reads: *"The system makes a visible relativistic jump of δ (where $\delta \in \mathbb{Z}$), and the continuing, unobserved future stream will satisfy property ϕ."* Because this evaluates future observation rather than past history, it serves as the formal logical engine for bisimulation and state-folding.
- $\mu X.\ \Phi(X)$ / $\nu X.\ \Phi(X)$ **(Least & Greatest Fixpoints):** The primitives for recursive behaviour. μ drives finite, bottom-up induction from baseline ignorance (reachability), while ν defines an endlessly repeating, top-down coinductive stream (persistence).

Example: Computing the Hanoi Parity Invariant

We can now express our hypothesis for the Towers of Hanoi in DTLF.

We co-induce an idealised invariant in DTLF where the smallest disk flawlessly cycles one peg to the left (-1) forever.

1. Proposing the Idealised Rule: We propose this property as our Greatest Fixpoint:

$$ParityInvariant \equiv \nu X.\ \langle -1 \rangle X$$

2. The Reality: However, the puzzle generates a stream that forces a forward jump of $+2$ from Peg_1 back to Peg_3. The observable bounds are:

$$Observed \equiv \nu X.\ (\langle -1 \rangle X \vee \langle +2 \rangle X)$$

3. Syntactic Bisimulation Equivalence: Because the topological futures of $+2$ and -1 land on the identical peg, we apply our syntactic equivalence ($\langle +2 \rangle \phi \equiv \langle -1 \rangle \phi$). This cleanly collapses the stream into our idealised property:

$$\nu X.\ (\langle -1 \rangle X \vee \langle +2 \rangle X)\ \equiv\ \nu X.\ \langle -1 \rangle X$$

Spatial Completeness & The Static Limitation: R-DTLF

The mathematical power of vanilla DTLF lies in **Spatial Completeness** (rooted in Stone Duality).

- **The Guarantee:** It proves a 1-to-1 isomorphism between the algebraic, step-by-step **Syntax** ($\phi \vdash \psi$) and the continuous, geometric **Semantics** ($[\![\phi]\!] \subseteq [\![\psi]\!]$).
- **Catching Limits:** Computations resolve when the directed path of finite evidence mathematically catches (satisfies) an infinite continuous topological limit (a Supremum).

Vanilla DTLF describes a single, *static* universe.

- To model the *dynamic* aspects of the Relativity Axioms, the Reference Frame itself must dynamically shift, so we use a relativistic version of DTLF, termed R-DTLF.

Compilation into SSAs & Topological Booleanization

R-DTLF does not discard classical logic; it functionally *generates* it at the semantic limits.

Theorem III: Topological Booleanization (Compiling SSAs)

Upon semantic realisation, the continuous, intuitionistic trace generated by RDo can be losslessly compiled into a discrete, classical First-Order representation via the **Lawvere-Tierney Double Negation ($\neg\neg$) Topology**.

Proof Sketch: Topos theory guarantees that inside every fluid, intuitionistic space exists a rigid **Boolean sub-topos**. Projecting onto this classical space preserves First-Order quantification while safely restoring the Law of Excluded Middle and hard negation.

Compilation into SSAs & Topological Booleanization (cont'd)

- **Compiling Successor State Axioms (SSAs):** This geometric compression is the categorical homologue to Ray Reiter's *Explanation Closure*. The Relativistic approach acts as a logic compiler—mathematically closing the open intuitionistic trace, and synthesising the discovered invariant directly into First-Order rules.
- **Hybrid Execution:** Because of this formal theorem, we can securely export newly learned R-DTLF invariants directly into standard classical Basic Action Theories (BATs).

This enables a powerful hybrid architecture: R-DTLF handles complex, infinite structural learning, and hands the compiled, closed-world rules off to legacy First-Order solvers (like Golog) for rapid execution.

Updated Towers of Hanoi Poss & SSA:

$$
\begin{aligned}
Poss(move(d, p_{from}, p_{to}), s) \equiv\ & Disk(d) \wedge (p_{from} \neq p_{to}) \wedge OnPeg(d, p_{from}, s) \wedge \\
& \neg \exists d' \Big(Disk(d') \wedge Smaller(d', d) \wedge \\
& (OnPeg(d', p_{from}, s) \vee OnPeg(d', p_{to}, s)) \Big) \\
& \wedge \big(d = D_1 \leftrightarrow Parity(s)\big) \\
& \wedge \big(d = D_1 \rightarrow p_{to} \equiv (p_{from} + \Delta) \pmod 3\big)
\end{aligned}
$$

$$
\begin{aligned}
OnPeg(d, p, do(a, s)) \equiv\ & \Big([\exists p_{from}\ (a = move(d, p_{from}, p))] \\
& \wedge \big(d = D_1 \rightarrow p \equiv (p_{from} + \Delta) \pmod 3\big) \Big) \vee \\
& OnPeg(d, p, s) \wedge \neg [\exists p_{to}\ (a = move(d, p, p_{to}))]
\end{aligned}
$$

Conclusion

Conclusion

Feature	Standard Situation Calculus (Induction)	Synthetic Relativity (R-DTLF) (Coinduction)
Objective	Historical Reachability / Action Execution	Property Persistence / Self-transcending reasoning
Conceptual Math	Least Fixpoint (LFP)	Greatest Fixpoint (GFP) & Least Fixpoint LFP
Logical Foundation	Second-Order Induction (Algebraic)	Co-induction (Coalgebraic) + Induction (Algebraic)
Syntactic Transformation	Transformation into First-Order Logic (SSAs)	Transformation to coinduced properties then compiled to SSAs
Computation	Computed as LFP in the Do modality (Golog)	Computed in the DTLF + Do modality (Golog)
Resolution	First-Order Theorem Proving	Bisimulation Equivalence (Quotients) + First-Order Theorem Proving

Related Work

Related Work

1. Action Theories & Fixpoint Verification

- **Model Checking in the Situation Calculus with the μ calculus:** [De Giacomo et al.]

By quotienting infinite state spaces via bisimulation into finite abstractions, they enabled the formal verification of dynamic, non-terminating systems using standard fixpoint model checking.

2. Categorical Logic & Observational Duality

- **Domain Theory in Logical Form (DTLF):** [Abramsky, 1991] established the foundational Stone Duality connecting denotational semantics (continuous topological domains) to program logic (finite observable properties).

Modern Coalgebraic Logic extends this duality to state-transition systems, directly mirroring our framework's state-collapse mechanism.

Related Work (continued)

3. Bounding Infinite Search & Forcing Convergence

- **Abstract Interpretation & Widening:** To guarantee least fixpoint convergence in infinite lattices, [Cousot & Cousot, 1977] introduced the **widening operator (∇)**.

Similar to our topological limit (OD_0), widening mathematically extrapolates a diverging inductive sequence, forcing it to snap to a safe, stable limit in finite time.

- **Inverse Entailment for Inductive Logic Programming (ILP):** [Muggleton 1995] avoids generating infinite relative least general generalisations (RLGGs).

Inverse Entailment bounds the inductive search strictly between the empty clause (most general) and a finite bottom clause (most specific), grounding background knowledge relative to finite examples.

- **Equational Anti-Unification:** [Cerna & Kutsia, 2023] In modern program synthesis, anti-unification is applied over E-graphs to efficiently extract common programmatic substructures and compress code without diverging into infinite logical spaces.

Future Work

Future Work: Formalising Relativistic Invariance

In physical relativity, shifting reference frames mathematically preserves the fundamental 4D spacetime interval across all frames ($\Delta s^2 = \Delta s'^2$):
$\Delta s^2 = (c\Delta t)^2 - (\Delta x^2 + \Delta y^2 + \Delta z^2)$.
We hypothesise a categorical equivalent for our semantic limits.

Conjecture: Relativistic Invariance of the Semantic Interval

The Hypothesis: Under the **Synthetic Transformation**, the semantic integrity of the realised limit (O) is structurally preserved across the categorical Change of Base.

Proposed Formalisation: By modelling the reference frame shift as a **Geometric Morphism**, we invoke its inverse image functor (f^*). While a Change of Base does not universally preserve all classical logic, Topos theory mathematically guarantees that f^* strictly preserves **Geometric Logic** (finite limits and arbitrary colimits). Because our inductive trace is built purely on these observable primitives, its geometric integrity is mathematically guaranteed to survive the shift.

- **Next Steps:** Formalising the Beck-Chevalley boundary conditions required to ensure seamless quantifier preservation ($\forall$, $\exists$) when the system mathematically synthesises higher-order subproblems.

Self-Transcending, Theory-Driven Generalisation

When the converged limit shifts to become the new Reference Frame (S_f) via the **Synthetic Transformation**, it acts natively as a global Greatest Fixpoint (ν) and the coinduced property is permanently added to the reference theory. The system treats **generalisation** not as a heuristic search, but as a rigorous categorical operation over S_f, governed by the foundational duality of fibrations and adjunctions:

- **1. Change of Base (The Fibrational Shift)**
  - Logical properties form a *Total Category* fibered over a *Base Category* of states and reference frames.
  - The system performs a categorical **Change of Base**, pulling the fully realised semantic space (ODo) down to become the new geometric foundation for future inductive proofs (RDo). Complex, inductively proven histories are safely compressed into the atomic geometry of the new baseline frame.

Self-Transcending, Theory-Driven Generalisation (cont'd)

- **2. Adjoint Duality (Swapping the Problem)**
  - Generalising a safe invariant is structurally the dual to deductive planning (a Categorical Adjunction: $L \dashv R$).
  - *Planning (Left Adjoint, L):* Inductive, bottom-up synthesis of a finite trace (Least Fixpoint, μ). Left adjoints natively preserve colimits.
  - *Generalising (Right Adjoint, R):* Coinductive, top-down projection of an infinite bound (Greatest Fixpoint, ν). Right adjoints natively preserve limits. Finite local observations mathematically push across the adjunction to rigorously force the coinductive proposal.

Self-Transcending, Theory-Driven Generalisation (cont'd)

- **3. Universal Properties (Terminal Coalgebras)**
  - The system requires no external human engineering templates or artificial "Syntax-Guided Synthesis" (SyGuS) biases.
  - Because S_f natively encodes the geometric symmetries of the domain, the optimal coinductive generalisation automatically emerges as a **Universal Property** (specifically, a *Terminal Coalgebra*).
  - **Stone Duality** mathematically guarantees that the internal logical syntax will organically fold to bound the state space's innate topology—achieving true, self-transcendence.

Synthetic Relativity versus Situation Calculus versus GLB

Aspect	*Synthetic Relativity (R-DTLF)*	*Situation Calculus (BATs/SSAs) $+$ variants of μ-calculus*	*Bimodal Provability logic (GLB)*
Modalities	RDo, ODo & PDo	PDo, $\langle a \rangle \phi$ (possibility), $[a]\phi$ (necessity)	$\Box$, $\Box_\omega$
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Completeness?	Spatially complete (continuous geometry) and Turing complete (universal computation via recursive domains); Arithmetically incomplete (triggers Gödel's incompleteness).	Turing complete (can compute anything), but neither spatially complete (no continuous geometry) nor arithmetically complete (subject to Gödel's incompleteness).	Arithmetically complete (flawlessly models the bounds of provability), but neither Turing complete (strictly decidable, cannot compute loops) nor spatially complete.
Other	Algebraic compactness. Kleene's Realizability & BHK (Brouwer-Heyting-Kolmogorov) Interpretation.	Regression Theorem (solves the frame problem). Fixpoint Semantics & Knaster-Tarski Theorem.	Japaridze's Arithmetical Completeness Theorem. Löb's Theorem & Well-founded Kripke Semantics.